CS57300
PURDUE UNIVERSITY

DATA MINING

ANNOUNCEMENT

- Project guideline will be out soon!
  - Form a team of 2-5 students to work on a data-mining related course project!
  - Connecting the course project to your research is encouraged!
- Project proposal: Due by Friday, September 22 (11:59pm)

OVERVIEW

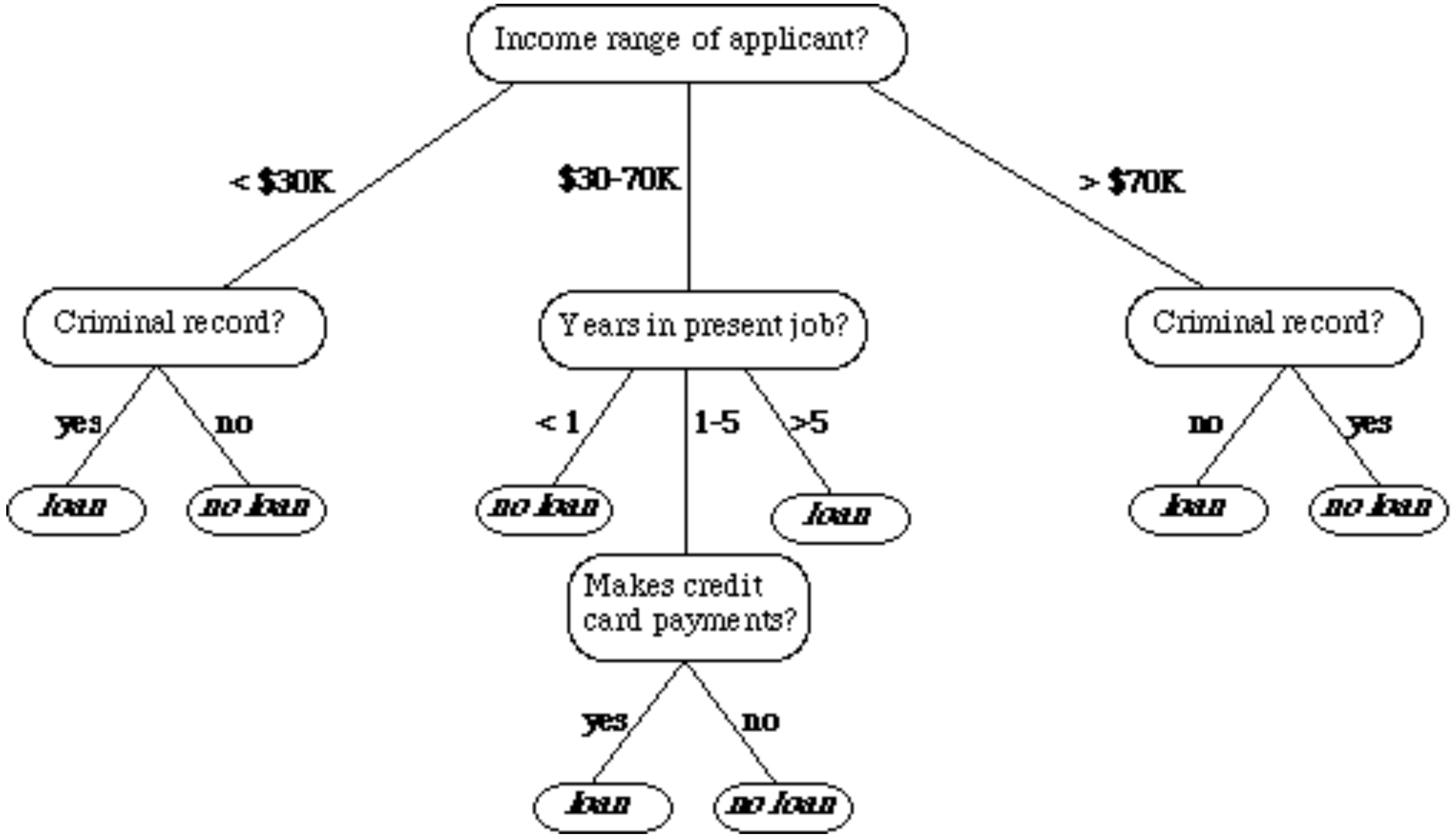
- Task specification
- **Knowledge representation**
- Learning technique
    - Search + scoring
- Prediction and/or interpretation

KNOWLEDGE REPRESENTATION

- Underlying structure of the model or patterns that we seek from the data
  - Specifies the models/patterns that could be returned as the results of the data mining algorithm
  - Defines space of possible models/patterns for algorithm to search over

KNOWLEDGE REPRESENTATION EXAMPLE: PREDICTIVE MODELING

- If-then rule
  - *If* (personal income > $70k) AND (criminal record = 'no'), *then* loan=yes
- Decision tree
  - Each node corresponds to an attribute
  - Each leaf is a class label

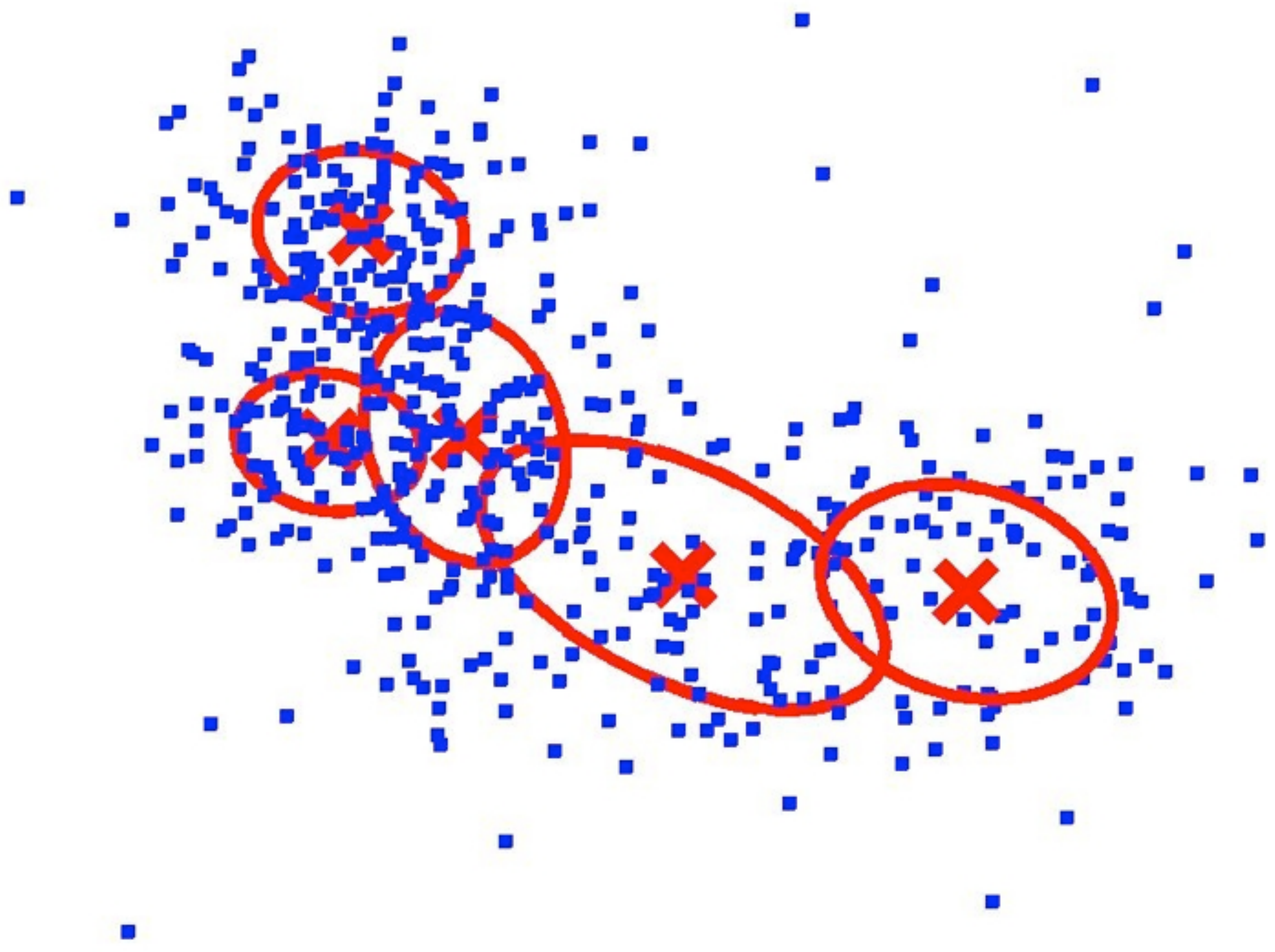


KNOWLEDGE REPRESENTATION EXAMPLE: PREDICTIVE MODELING

- Conditional probability distributions (i.e., P(Y|$\boldsymbol{X}$))
  - Logistic regression: $log\frac{P(Y=1|\boldsymbol{x})}{1-P(Y=1|\boldsymbol{x})}=\beta_0+\boldsymbol{\beta x}$
  - Model the log-odds as a linear combination of predictors
- Linear regression
  - $y=\beta_1 x_1+\beta_2 x_2 \ldots+\beta_0$
  - y is the response variable, $\boldsymbol{x}$ is the predictor variable

KNOWLEDGE REPRESENTATION EXAMPLE: DESCRIPTIVE MODELING

- Mixture model: Instances represented as a weighted combination of mixture distributions



$$f(x) = \sum_{k=1}^{K} w_k f_k(x;\theta)$$

probability of observing x

likelihood of x being generated from cluster k

likelihood of point belonging to cluster k

KNOWLEDGE REPRESENTATION EXAMPLE: PATTERN DISCOVERY

- Association rules
  - $I = \{i_1, i_2, \ldots, i_n\}$ is a set of n items
  - $T = \{t_1, t_2, \ldots, t_m\}$ is a set of m transactions
  - An association rule has the form $X \rightarrow Y$, where X and Y are subsets of I, which means if items in X appear in a transaction, then items in Y are likely to appear in that transaction
    - E.g., {beer} → {diaper}; {bread} → {milk}

Example database with 5 transactions and 5 items

transaction ID	milk	bread	butter	beer	diapers
1	1	1	0	0	0
2	0	0	1	0	0
3	0	0	0	1	1
4	1	1	1	0	0
5	0	1	0	0	0

OVERVIEW

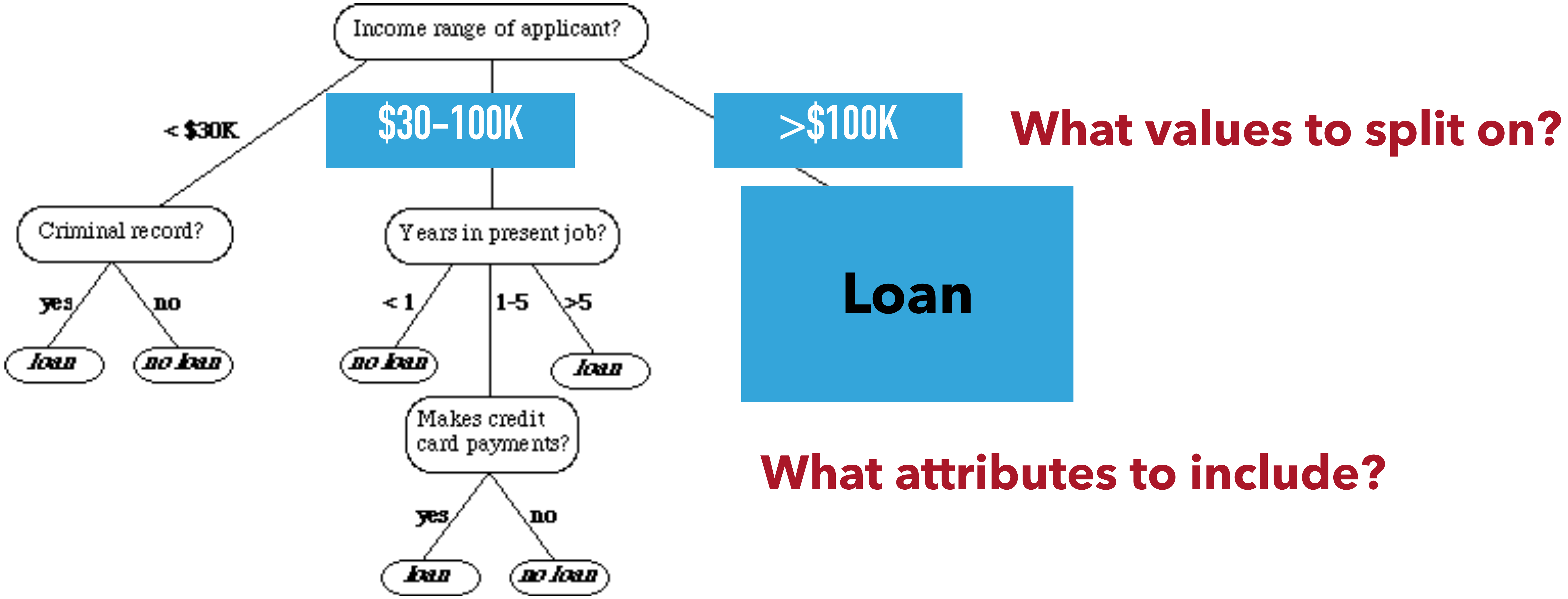
- Task specification
- Knowledge representation
- **Learning technique**
  - Search + scoring
- Prediction and/or interpretation

LEARNING TECHNIQUE

- Method to construct model or patterns from data
- **Model space**
  - Choice of knowledge representation defines a set of possible models or patterns
- **Scoring function**
  - Associates a numerical value (score) with each member of the set of models/patterns
- **Search technique**
  - Defines a method for generating members of the set of models/patterns, determining their score, and identifying the ones with the "best" score

MODEL SPACE

- Defined by the choice of knowledge representation
- Decision tree:



MODEL PARAMETERS AND STRUCTURE

- Models have both **parameters** and **structure**
- **Parameters:**
  - Feature values in classification tree
  - Coefficients in regression model
  - Probability estimates in graphical model
- **Structure:**
  - Nodes in classification tree
  - Variables in regression model
  - Edges in graphical model

SCORING FUNCTION

- A numeric score assigned to each possible model in a search space, **given a reference/input dataset**
  - Used to judge the quality of a particular model for the domain
- Score function are **statistics**—estimates of a population parameter based on a sample of data
- Examples:
  - Misclassification
  - Squared error
  - Likelihood

PARAMETER ESTIMATION VS. STRUCTURE LEARNING

- **Parameters:**
  - Feature values in classification tree
  - Coefficients in regression model
  - Probability estimates in graphical model

Search: Convex/smooth optimization techniques

- **Structure:**
  - Nodes in classification tree
  - Variables in regression model
  - Edges in graphical model

Search: Heuristic approaches for combinatorial optimization

EXAMPLE LEARNING PROBLEM

Task: Devise a rule to classify items based on the attribute ***X***

Knowledge representation:

If-then rules

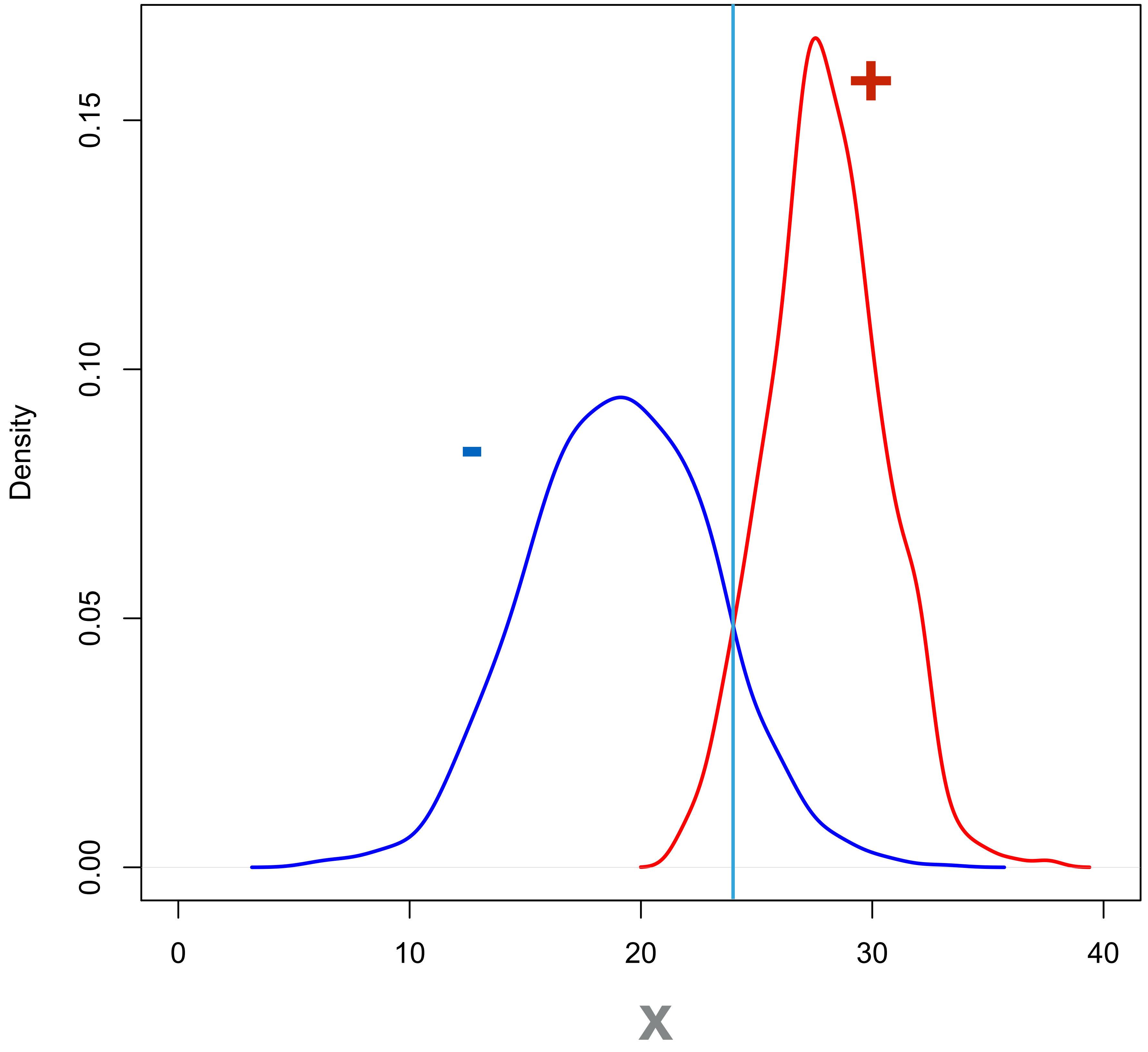
Example rule:

If x > 24 then +

Else -

What is the model space?

All possible thresholds



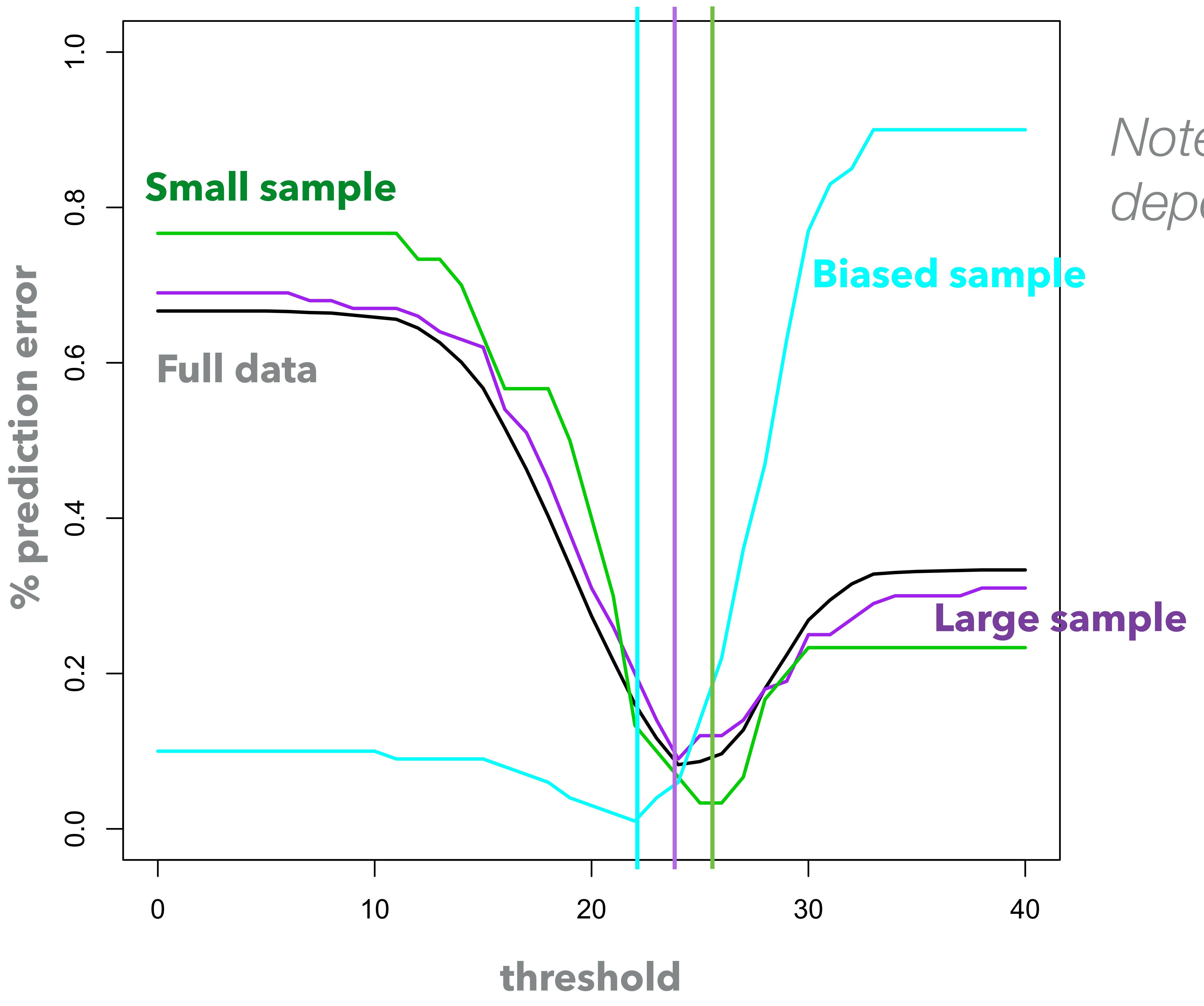
What score function?

Prediction error rate

SCORE FUNCTION OVER MODEL SPACE

Search procedure?

Try all thresholds, select one with lowest score



*Note: learning result depends on **data***